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PRODUCT NAME

NP Ferrous Fumarate 200mg Tablet

COMPOSITION

Ferrous Fumarate

200mg/tablet

PRODUCT DESCRIPTION

A brown coloured, round shaped film coated tablet

PHARMACODYNAMICS

Iron and iron salts should be given for the treatment or prophylaxis of iron deficiency anaemias. Iron is an essential component in the physiological formation of haemoglobin, adequate amounts of which are necessary for effective erythropoiesis and the resultant oxygen transport capacity of the blood. A similar function is provided by iron in myoglobin production. Iron also serves as a co-factor of several essential enzymes. Ferrous fumarate is an easily absorbed source of iron for replacement therapy. It is a salt of ferrous iron with an organic acid and is less irritant to the gastro-intestinal tract than salts with inorganic acids.

PHARMACOKINETICS

Absorption

In the acid conditions of the gastric contents, ferrous fumarate is dissociated and ferrous ions are liberated. These irons are absorbed in the proximal portion of the duodenum.

The ferrous iron absorbed by the mucosal cells of the duodenum is oxidized to the ferric form, and this is bound to a protein to form ferritin.

Absorption is most efficient when iron is ingested in its ferrous rather than ferric form on empty stomach when taken with food, the amount of iron absorbed may be reduced by ½ to 1/3 as when taken on empty stomach.

Distribution

Ferritin in the mucosal cells releases iron into the blood, where it is bound to transferrin and passed into the iron stores - liver, spleen, and bone marrow.

Protein binding: Very High

Haemoglobin – High

Myoglobin, enzymes and Transferrin – Low

Ferritin and Hemosiderin – Low

These stores are a reserve of iron for synthesis of hemoglobin, myoglobin, and iron containing enzymes.

Excretion:

Iron is lost from the body through loss of cells in urine, faeces, hair, skin, sputum, nails, and mucosal cells, and through blood loss.

Ferrous fumarate has the same pattern of absorption and excretion as dietary iron.

INDICATION

Prophylaxis and treatment of iron deficiency anemia.

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RECOMMENDED DOSAGE

Adult:

Treatment - 1 tablet two to three times a day;

Prophylaxis - 1 tablet once or twice a day.

ROUTE OF ADMINISTRATION

Oral.

CONTRAINDICATIONS

Ferrous fumarate is contraindicated in patients with:

- Hemochromatosis
- Hemosiderosis
- Anemic conditions not due to iron deficiency
- Sensitivity to iron
- Porphyria cutanea tarda.

WARNING AND PRECAUTIONS

Patients with intestinal strictures and diverticular disease. May worsen diarrhoea in patients with inflammatory bowel disease. May cause constipation and faecal impaction in elderly. Avoid prolonged admin (>6 mth) except in patients with continued bleeding, menorrhagia or repeated pregnancies. Not for routine use in treatment of haemolytic anaemia unless an iron-deficient state exists. Parenteral iron should not be used concurrently with oral iron treatment. Avoid use in patients receiving repeated blood transfusions.

Pregnancy.

In the first trimester of pregnancy, adequate iron intake is usually obtained from a proper diet; however, in the second and third trimesters when iron deficiency is more prevalent because of greatly increased requirements, iron supplements often are necessary. Problems in human have not been documented with orally ingested iron with intake of daily recommended amount.

Breast feeding

Problems in human have not been documented with orally ingested iron with intake of daily recommended amount.

Geriatrics

Some geriatric patients may require a larger than usual daily ingestion of iron deficiency because their ability to absorb iron has been diminished by reduced gastric secretions and achlorhydria.

DRUG INTERACTIONS

Iron reduces the absorption of penicillamine, bisphosphonates, ciprofloxacin, entacapone, levodopa, levofloxacin, levothyroxine (thyroxine) (give at least 2 hours apart), moxifloxacin, mycophenolate, norfloxacin, ofloxacin, zinc. Absorption of both iron and antibiotic may be reduced if NP Ferrous Fumarate is given with tetracycline.

Absorption of oral iron is reduced by calcium salts, Magnesium salts (as magnesium trisilicate), Trientine.

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Chloramphenicol delays plasma iron clearance, incorporation of iron into red blood cells and interferes with erythropoiesis. Some inhibition of iron absorption may occur if it is taken with cholestyramine, tea, eggs or milk.

Avoid concomitant use of iron with Dimercaprol.

Oral iron antagonises hypotensive effect of methyldopa.

ADVERSE EFFECTS

GI: Nausea, vomiting, GI discomfort, epigastric pain, constipation or diarrhoea, anorexia, dark stool, dental discolouration, heartburn.

Genitourinary: Urine discolouration.

OVERDOSE AND TREATMENT

Symptoms: Nausea, vomiting, abdominal pain, diarrhoea. In more serious cases, hypoperfusion (cool peripheries and hypotension), metabolic acidosis, recurrence of vomiting, GI bleeding, shock, hepatocellular necrosis, jaundice, hypoglycaemia, encephalopathy, renal failure, and rarely, gastric scarring leading to stricture or pyloric stenosis.

Treatment: Supportive and symptomatic treatment. Whole bowel irrigation may be considered. If metabolic acidosis persists despite correction of hypoxia and adequate fluid resuscitation, an initial dose of Na bicarbonate 50 mmol may be given and repeated as necessary. Consider the use of desferrioxamine, if the patient is symptomatic, serum Fe concentration is between 3-5 mg/L and still rising.

STORAGE CONDITIONS

Store below 30°C.

Protect from light.

PRESENTATION

PVC/Aluminium blister of 10 tablets.

Pack Size: 30s, 50's, 100's, 300's, 500's and 1000's.

MANUFACTURED AND PRODUCT REGISTRATION HOLDER

Noripharma Sdn Bhd.,
Lot 5030 Jalan Teratai,
5½ Mile Off Jalan Meru,
41050 Klang, Selangor.

DATE OF REVISION

October 2018